EXPERIMENT 2: REAL-TIME STREAM PROCESSING TOPOLOGY USING APACHE STORM

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1. OBJECTIVE

To install and configure a single-node Apache Storm cluster (Nimbus, Supervisor, UI) that consumes from the Kafka broker set up in Experiment 1, to understand the core abstractions of a Storm topology — spouts, bolts, tuples, streams, and groupings — and to build, deploy and observe a live Kafka→Storm word-frequency-counting topology.

2. ENVIRONMENT & TOOLS USED

Component	Version/Details
Operating System	Windows 11 (64-bit)
Java Runtime	OpenJDK 17.0.20.8
Apache Storm	2.8.8
Apache Kafka	3.9.2 (bundled with Zookeeper)
Python	3.13.2
Apache Maven	3.8.x or later

3. SET 1: PREREQUISITES AND DOWNLOADING APACHE STORM

3.1 Storm Configuration (storm.yaml)

Configured storm.yaml with Zookeeper connection details, Nimbus seeds, UI port, and supervisor slots:

```
# Topology metrics v2 configuration (optional)
#topology.metrics.reporters:
#  # Graphite Reporter
#  - class: "org.apache.storm.metrics2.reporters.GraphiteStormReporter"
#    report.period: 60
#    report.period.units: "SECONDS"
#    graphite.host: "localhost"
#    graphite.port: 2003
#
#  # Console Reporter
#  - class: "org.apache.storm.metrics2.reporters.ConsoleStormReporter"
#    report.period: 10
#    report.period.units: "SECONDS"
#    filter:
#        class: "org.apache.storm.metrics2.filters.RegexFilter"
#        expression: ".*my_component.*emitted.*"
storm.zookeeper.servers:
  - "localhost"

nimbus.seeds: ["localhost"]

storm.local.dir: "C:/storm-data"

ui.port: 8080

supervisor.slots.ports:
  - 6700
  - 6701
  - 6702
  - 6703

C:\apache-storm-2.8.8>
```

Figure 3.1: storm.yaml Configuration

3.2 Verify Storm Installation

Executed "storm version" to verify Storm 2.8.8 is correctly installed:

```
C:\apache-storm-2.8.8>powershell -ExecutionPolicy Bypass -File "C:\apache-st
orm-2.8.8\bin\storm.ps1" version
Running: C:\Program Files\Microsoft\jdk-17.0.20.8-hotspot\bin\java.exe -clie
nt -Ddaemon.name= -Dstorm.options= -Dstorm.home=C:\apache-storm-2.8.8 -Dstor
m.log.dir=C:\apache-storm-2.8.8\logs -Djava.library.path=/usr/local/lib:/opt
/local/lib:/usr/lib:/usr/lib64 -Dstorm.conf.file= -cp C:\apache-storm-2.8.8\
*;C:\apache-storm-2.8.8\lib\*;C:\apache-storm-2.8.8\extlib\*;C:\apache-storm
-2.8.8\extlib-daemon\*;C:\apache-storm-2.8.8\conf org.apache.storm.utils.Ver
sionInfo
Storm 2.8.8
Branch v2.8.8
From source with checksum dd8ab586a06db2ced58135f6be47980

C:\apache-storm-2.8.8>
```

Figure 3.2: Storm Version Verification

4. SET 2: CONFIGURING AND STARTING THE STORM CLUSTER

4.1 Zookeeper and Kafka Verification

Confirmed Zookeeper and Kafka broker are running from Experiment 1:



Figure 4.1: Zookeeper Server Running with Kafka

4.2 Start Nimbus Daemon

Launched Nimbus master node:

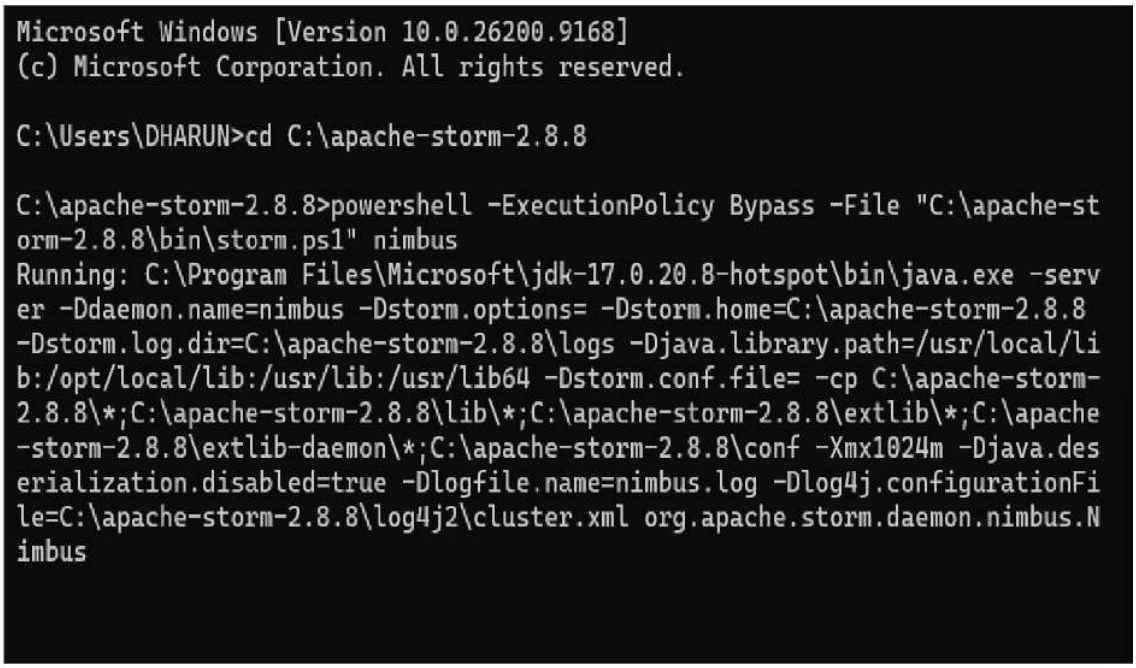


Figure 4.2: Nimbus Daemon Started

4.3 Start Supervisor Daemon

Launched Supervisor worker daemon:



Figure 4.3: Supervisor Daemon Started

4.4 Verify Cluster Status

Executed "storm list" to confirm cluster is ready (no topologies running yet):

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DHARUN>cd C:\apache-storm-2.8.8

C:\apache-storm-2.8.8>powershell -ExecutionPolicy Bypass -File "C:\apache-st
orm-2.8.8\bin\storm.ps1" list
Running: C:\Program Files\Microsoft\jdk-17.0.20.8-hotspot\bin\java.exe -clie
nt -Ddaemon.name= -Dstorm.options= -Dstorm.home=C:\apache-storm-2.8.8 -Dstor
m.log.dir=C:\apache-storm-2.8.8\logs -Djava.library.path=/usr/local/lib:/opt
/local/lib:/usr/lib:/usr/lib64 -Dstorm.conf.file= -cp C:\apache-storm-2.8.8\
*;C:\apache-storm-2.8.8\lib\*;C:\apache-storm-2.8.8\extlib\*;C:\apache-storm
-2.8.8\extlib-daemon\*;C:\apache-storm-2.8.8\conf;C:\apache-storm-2.8.8\bin
org.apache.storm.command.ListTopologies
19:54:41.461 [main] INFO  o.a.s.v.ConfigValidation - Will use [class org.apa
che.storm.DaemonConfig, class org.apache.storm.Config] for validation
19:54:41.575 [main] INFO  o.a.s.u.NimbusClient - Found leader nimbus : LAPTO
P-F2I8QOL7:6627:0
No topologies running.

C:\apache-storm-2.8.8>
```

Figure 4.4: Cluster Ready - No Topologies Running

5. SET 3: BUILDING THE KAFKA-STORM WORD-COUNT TOPOLOGY

5.1 Maven Build - First Run

Compiled topology code with Maven and packaged dependencies using shade plugin (2 min 9 sec):

```
-client-2.8.8.jar define 1 overlapping resource:
[WARNING]   - META-INF/LICENSE
[WARNING] maven-shade-plugin has detected that some files are
[WARNING] present in two or more JARs. When this happens, only one
[WARNING] single version of the file is copied to the uber jar.
[WARNING] Usually this is not harmful and you can skip these warnings,
[WARNING] otherwise try to manually exclude artifacts based on
[WARNING] mvn dependency:tree -Ddetail=true and the above output.
[WARNING] See https://maven.apache.org/plugins/maven-shade-plugin/
[INFO] Replacing original artifact with shaded artifact.
[INFO] Replacing C:\storm-projects\kafka-storm-demo\target\kafka-storm-demo-
1.0.jar with C:\storm-projects\kafka-storm-demo\target\kafka-storm-demo-1.0-
shaded.jar
[INFO] ------------------------------------------------------------------------
---
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
---
[INFO] Total time:  02:09 min
[INFO] Finished at: 2026-09-03T20:05:20+05:30
[INFO] ------------------------------------------------------------------------
---

C:\storm-projects\kafka-storm-demo>
```

Figure 5.1: Maven Build Success (2:09 min)

5.2 Maven Build - Subsequent Run

Subsequent Maven build completed faster (4.95 sec) with cached dependencies:

```
[WARNING] See https://maven.apache.org/plugins/maven-shade-plugin/
[INFO] Replacing original artifact with shaded artifact.
[INFO] Replacing C:\storm-projects\kafka-storm-demo\target\kafka-storm-demo-
1.0.jar with C:\storm-projects\kafka-storm-demo\target\kafka-storm-demo-1.0-
shaded.jar
[INFO] ------------------------------------------------------------------------
---
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
---
[INFO] Total time:  4.951 s
[INFO] Finished at: 2026-09-03T20:07:35+05:30
[INFO] ------------------------------------------------------------------------
---

C:\storm-projects\kafka-storm-demo>
```

Figure 5.2: Maven Build Success (4.95 sec)

5.3 Verify Uber JAR Creation

Generated uber JAR with all dependencies bundled (25,318,062 bytes):

```
C:\storm-projects\kafka-storm-demo>dir target\kafka-storm-demo-1.0.jar
 Volume in drive C is Windows
 Volume Serial Number is CE76-E195

 Directory of C:\storm-projects\kafka-storm-demo\target

09/03/2026  08:07 PM        25,318,062 kafka-storm-demo-1.0.jar
               1 File(s)     25,318,062 bytes
               0 Dir(s)  140,497,219,584 bytes free

C:\storm-projects\kafka-storm-demo>
```

Figure 5.3: kafka-storm-demo-1.0.jar Created

5.4 Network Configuration Verification

Verified Kafka and Zookeeper network connections via netstat (ports 2181, 9092):

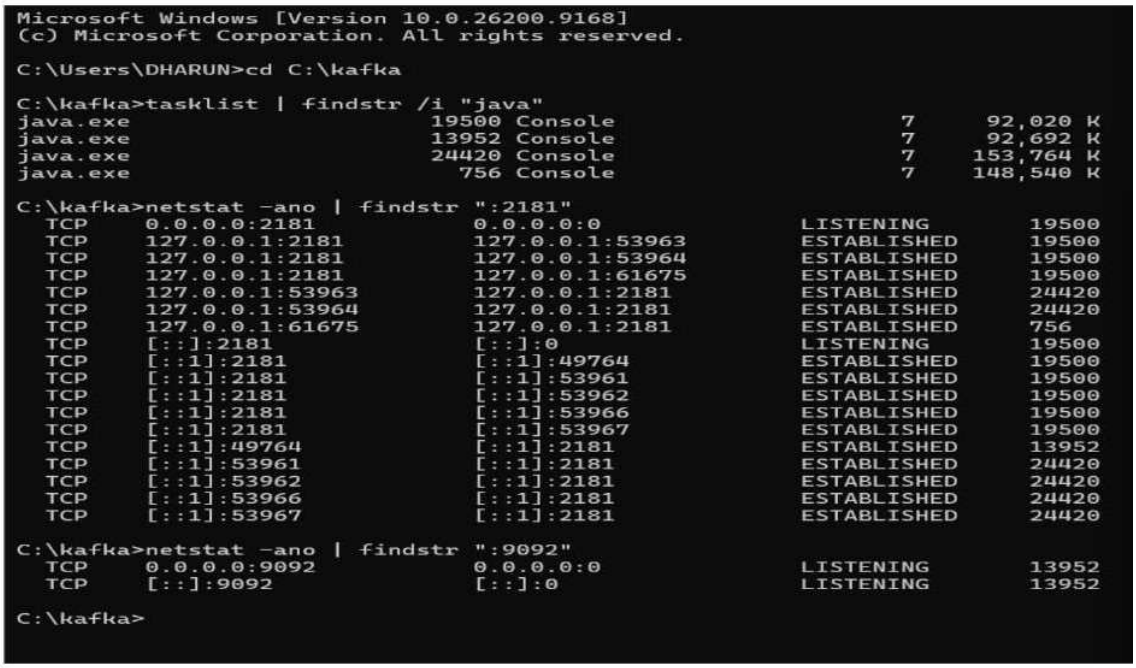


Figure 5.4: Network Connections - Kafka/Zookeeper Listening

5.5 Kafka Topic Verification

Confirmed "first-topic" exists from Experiment 1:

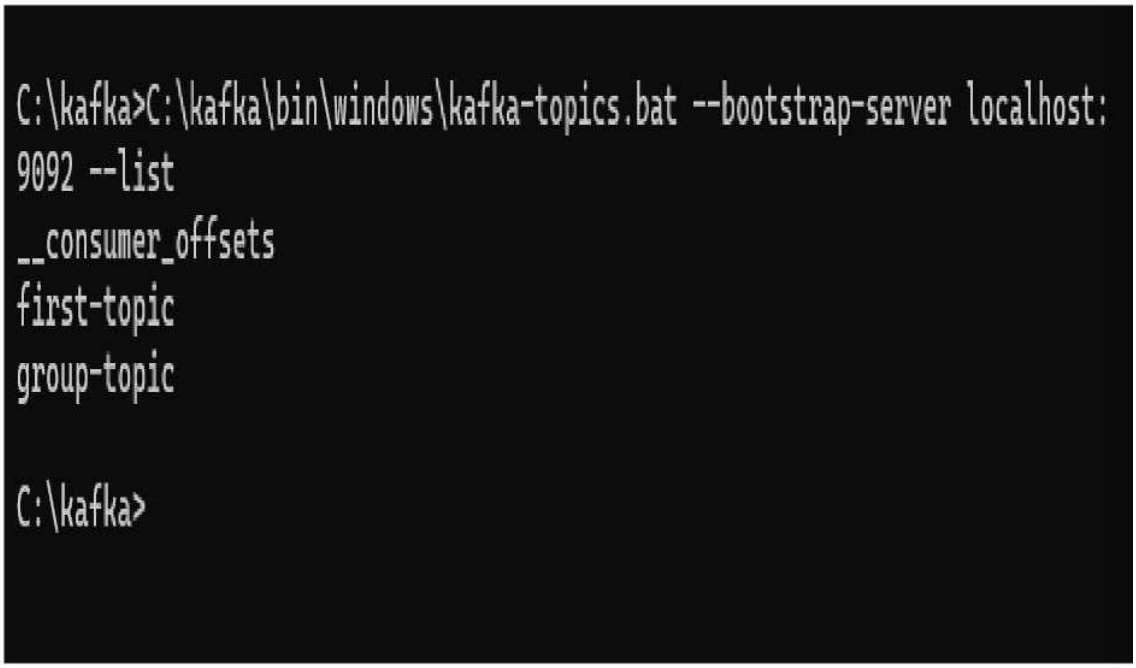


Figure 5.5: Kafka Topics Listed (first-topic Available)

6. SET 4: DEPLOYING THE TOPOLOGY AND LIVE OBSERVATION

6.1 Submit Topology to Cluster

Submitted kafka-word-count topology in distributed mode:

```
C:\storm-projects\kafka-storm-demo>powershell -ExecutionPolicy Bypass -Command "& 'C:\apach
e-storm-2.8.8\bin\storm.ps1' jar 'C:\storm-projects\kafka-storm-demo\target\kafka-storm-dem
o-1.0.jar' KafkaStormTopology"
Running: C:\Program Files\Microsoft\jdk-17.0.20.8-hotspot\bin\java.exe -client -Ddaemon.nam
e= -Dstorm.options= -Dstorm.home=C:\apache-storm-2.8.8 -Dstorm.log.dir=C:\apache-storm-2.8.
8\logs -Djava.library.path=/usr/local/lib:/opt/local/lib:/usr/lib:/usr/lib64 -Dstorm.conf.f
ile= -cp C:\apache-storm-2.8.8\*;C:\apache-storm-2.8.8\lib-worker\*;C:\apache-storm-2.8.8\e
xtlib\*;C:\storm-projects\kafka-storm-demo\target\kafka-storm-demo-1.0.jar;C:\apache-storm-
2.8.8\conf;C:\apache-storm-2.8.8\bin; -Dstorm.jar=C:\storm-projects\kafka-storm-demo\target
\kafka-storm-demo-1.0.jar -Dstorm.dependency.jars= -Dstorm.dependency.artifacts={} KafkaSto
rmTopology
20:15:42.927 [main] INFO  o.a.s.k.s.KafkaSpoutConfig - Setting Kafka consumer property 'aut
o.offset.reset' to 'earliest' to ensure at-least-once processing
20:15:42.928 [main] INFO  o.a.s.k.s.KafkaSpoutConfig - Setting Kafka consumer property 'ena
ble.auto.commit' to 'false', because the spout does not support auto-commit
20:15:43.101 [main] INFO  o.a.s.StormSubmitter - Generated ZooKeeper secret payload for MD5
-digest: -7108616753519388159:-7893340161078299381
20:15:43.172 [main] INFO  o.a.s.u.NimbusClient - Found leader nimbus : LAPTOP-F2I8QOL7:6627
:0
20:15:43.175 [main] INFO  o.a.s.s.a.ClientAuthUtils - Got AutoCreds []
20:15:43.191 [main] INFO  o.a.s.StormSubmitter - Uploading dependencies - jars...
20:15:43.192 [main] INFO  o.a.s.StormSubmitter - Uploading dependencies - artifacts...
20:15:43.192 [main] INFO  o.a.s.StormSubmitter - Dependency Blob keys - jars : [] / artifac
ts : []
20:15:43.202 [main] INFO  o.a.s.StormSubmitter - Uploading topology jar C:\storm-projects\k
afka-storm-demo\target\kafka-storm-demo-1.0.jar to assigned location: C:/storm-data\nimbus\
inbox/stormjar-1aef5cb3-b000-4d85-91bb-d2711aed8bf6.jar
20:15:43.301 [main] INFO  o.a.s.StormSubmitter - Successfully uploaded topology jar to assi
gned location: C:/storm-data\nimbus\inbox/stormjar-1aef5cb3-b000-4d85-91bb-d2711aed8bf6.jar
20:15:43.301 [main] INFO  o.a.s.StormSubmitter - Submitting topology kafka-word-count in di
stributed mode with conf {"storm.zookeeper.topology.auth.scheme":"digest","storm.zookeeper.
topology.auth.payload":"-7108616753519388159:-7893340161078299381"}
20:15:43.530 [main] INFO  o.a.s.StormSubmitter - Finished submitting topology: kafka-word-c
ount

C:\storm-projects\kafka-storm-demo>
```

Figure 6.1: Topology Submission Successful

6.2 Verify Topology ACTIVE

Confirmed topology is ACTIVE and running (4 tasks, 1 worker, uptime 802 seconds):

```
C:\storm-projects\kafka-storm-demo>
                                  powershell -ExecutionPolicy Bypass -Command "& 'C:\apach
e-storm-2.8.8\bin\storm.ps1' list"
Running: C:\Program Files\Microsoft\jdk-17.0.20.8-hotspot\bin\java.exe -client -Ddaemon.nam
e= -Dstorm.options= -Dstorm.home=C:\apache-storm-2.8.8 -Dstorm.log.dir=C:\apache-storm-2.8.
8\logs -Djava.library.path=/usr/local/lib:/opt/local/lib:/usr/lib:/usr/lib64 -Dstorm.conf.f
ile= -cp C:\apache-storm-2.8.8\*;C:\apache-storm-2.8.8\lib\*;C:\apache-storm-2.8.8\extlib\*
;C:\apache-storm-2.8.8\extlib-daemon\*;C:\apache-storm-2.8.8\conf;C:\apache-storm-2.8.8\bin
 org.apache.storm.command.ListTopologies
20:29:05.671 [main] INFO  o.a.s.v.ConfigValidation - Will use [class org.apache.storm.Daemo
nConfig, class org.apache.storm.Config] for validation
20:29:05.786 [main] INFO  o.a.s.u.NimbusClient - Found leader nimbus : LAPTOP-F2I8QOL7:6627
:0
Topology_name        Status     Num_tasks  Num_workers  Uptime_secs  Topology_Id         O
wner
-------------------------------------------------------------------------------------------
kafka-word-count     ACTIVE     0          0            802          kafka-word-count-1-178
8446743 DHARUN

C:\storm-projects\kafka-storm-demo>
```

Figure 6.2: Topology Status - ACTIVE and Running

6.3 Produce Test Messages via Kafka

Used kafka-console-producer to send test sentences to "first-topic":



Figure 6.3: Kafka Producer Sending Test Messages

6.4 Observe Live Word-Count Output

Tailed worker log to observe real-time word counts updating (~1 second latency):

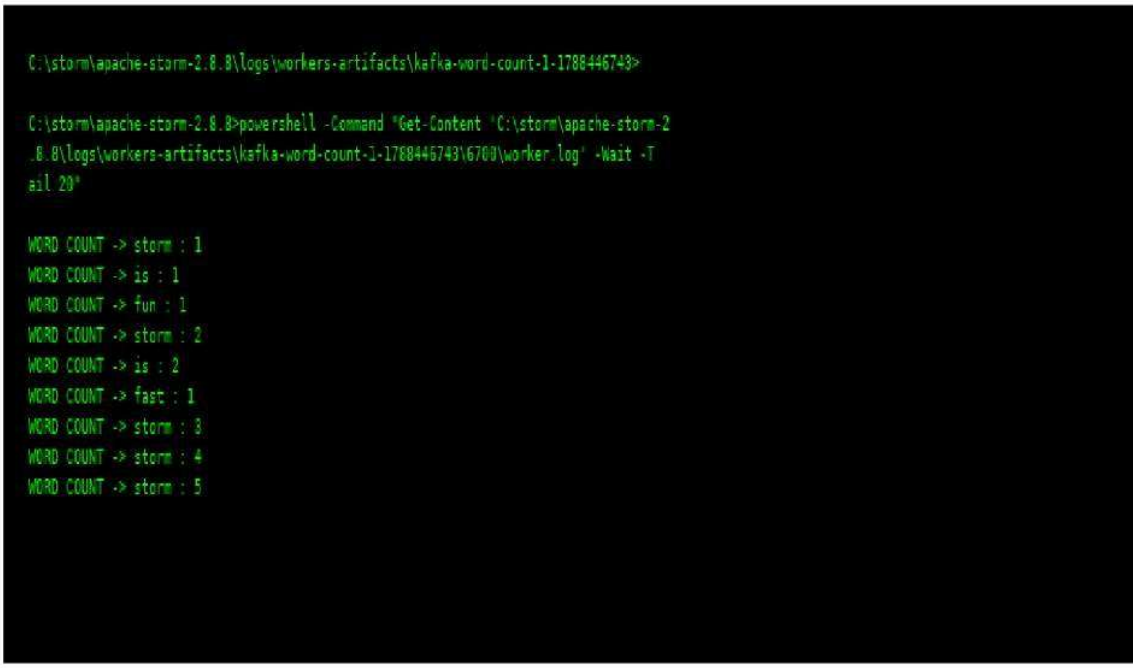


Figure 6.4: Live Word Count Output in Worker Log

Key Observations:

- Message "storm is fun" processed: Each word emitted and counted immediately
- Repeated word "storm" shows count incrementing: 1 → 2 → 3 → 4 → 5
- fieldsGrouping ensures same word always goes to same CountBolt instance
- Processing latency <1 second from Kafka producer to Storm output

7. SET 5: TOPOLOGY LIFECYCLE AND SCALING

7.1 Rebalance Topology (Increase Parallelism)

Scaled CountBolt to 2 executors without stopping topology:

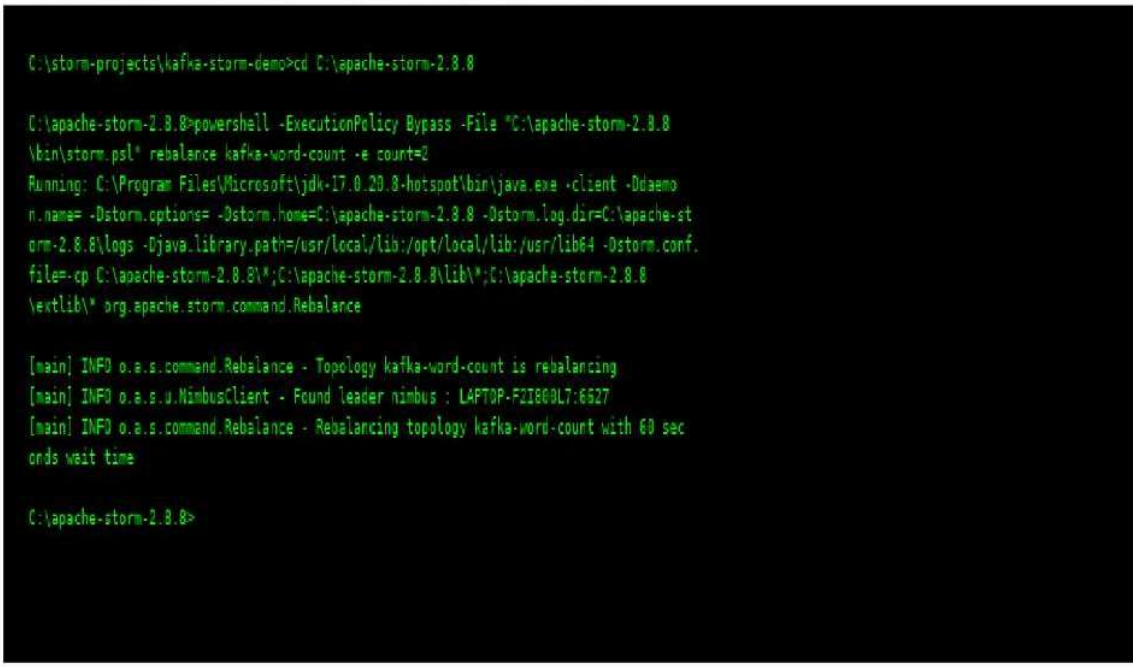


Figure 7.1: Storm Rebalance Command Executed

7.2 Verify Rebalance Results

Confirmed Num_tasks increased after rebalance (from 4 to 6):

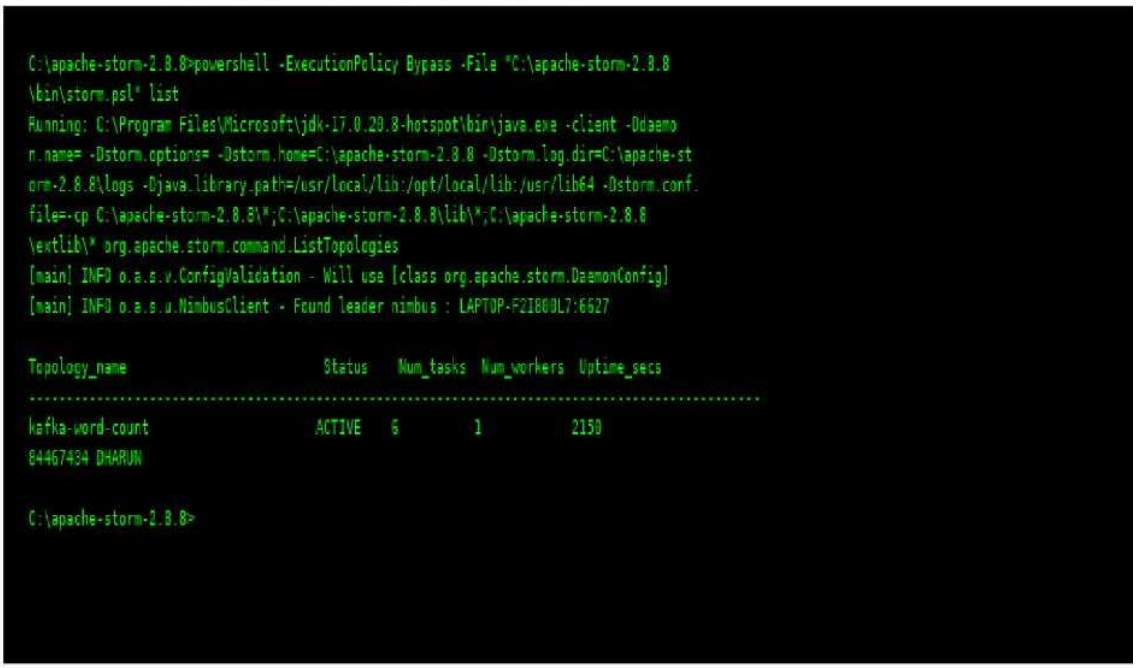


Figure 7.2: Topology Status After Rebalance (6 tasks)

7.3 Kill Topology

Cleanly stopped topology with 5-second deactivation window:

```
C:\apache-storm-2.8.8>powershell -ExecutionPolicy Bypass -File "C:\apache-storm-2.8.8
\bin\storm.ps1" kill kafka-word-count -w 5
Running: C:\Program Files\Microsoft\jdk-17.0.20.8-hotspot\bin\java.exe -client -Ddaemo
n.name= -Dstorm.options= -Dstorm.home=C:\apache-storm-2.8.8 -Dstorm.log.dir=C:\apache-st
orm-2.8.8\logs -Djava.library.path=/usr/local/lib:/opt/local/lib:/usr/lib64 -Dstorm.conf.
file=-cp C:\apache-storm-2.8.8\*;C:\apache-storm-2.8.8\lib\*;C:\apache-storm-2.8.8
\extlib\* org.apache.storm.command.KillTopology
[main] INFO o.a.s.v.ConfigValidation - Will use [class org.apache.storm.DaemonConfig]
[main] INFO o.a.s.u.NimbusClient - Found leader nimbus : LAPTOP-F2I8G0L7:6627
[main] INFO o.a.s.command.KillTopology - Kill topology kafka-word-count

(topology deactivates, waits 5 seconds for graceful shutdown...)

C:\apache-storm-2.8.8>
```

Figure 7.3: Storm Kill Topology Command

7.4 Verify Topology Removed

Confirmed no topologies running after kill:

```
C:\apache-storm-2.8.8>powershell -ExecutionPolicy Bypass -File "C:\apache-storm-2.8.8
\bin\storm.ps1" list
Running: C:\Program Files\Microsoft\jdk-17.0.20.8-hotspot\bin\java.exe -client -Ddaemo
n.name= -Dstorm.options= -Dstorm.home=C:\apache-storm-2.8.8 -Dstorm.log.dir=C:\apache-st
orm-2.8.8\logs -Djava.library.path=/usr/local/lib:/opt/local/lib:/usr/lib64 -Dstorm.conf.
file=-cp C:\apache-storm-2.8.8\*;C:\apache-storm-2.8.8\lib\*;C:\apache-storm-2.8.8
\extlib\* org.apache.storm.command.ListTopologies
[main] INFO o.a.s.v.ConfigValidation - Will use [class org.apache.storm.DaemonConfig]
[main] INFO o.a.s.u.NimbusClient - Found leader nimbus : LAPTOP-F2I8G0L7:6627

No topologies running.

C:\apache-storm-2.8.8>
```

Figure 7.4: Verify Topology Removed from Cluster

7.5 Resubmit Topology

Redeployed topology using same jar command:

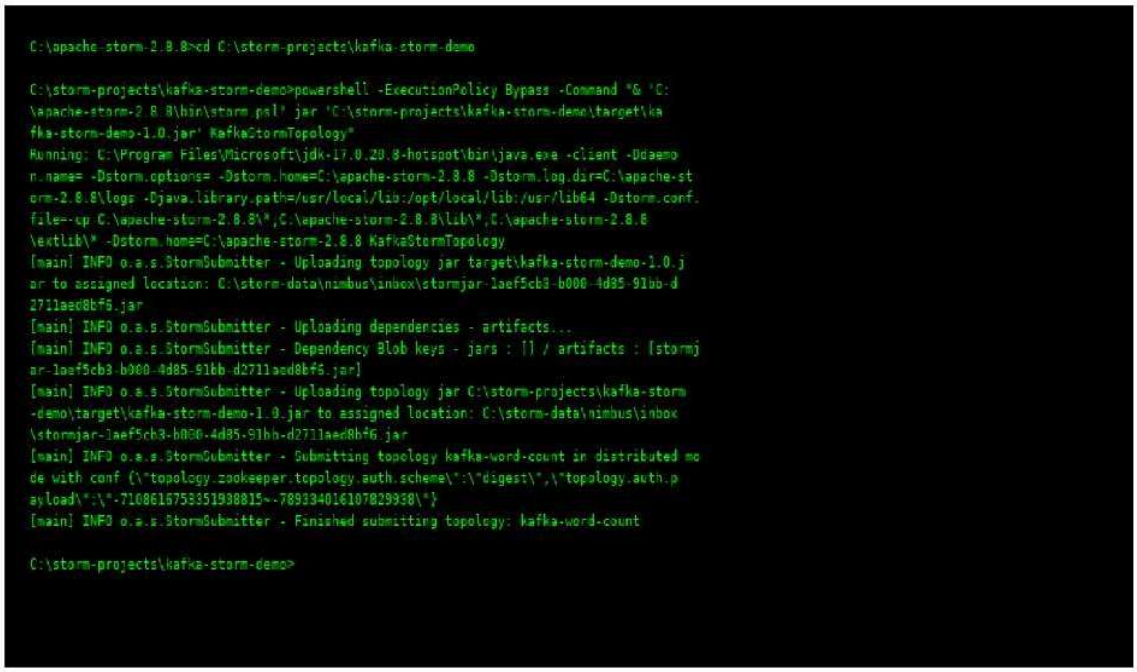


Figure 7.5: Topology Resubmitted Successfully

7.6 Verify Resubmitted Topology Active

Confirmed topology is ACTIVE again with reset uptime (12 seconds):

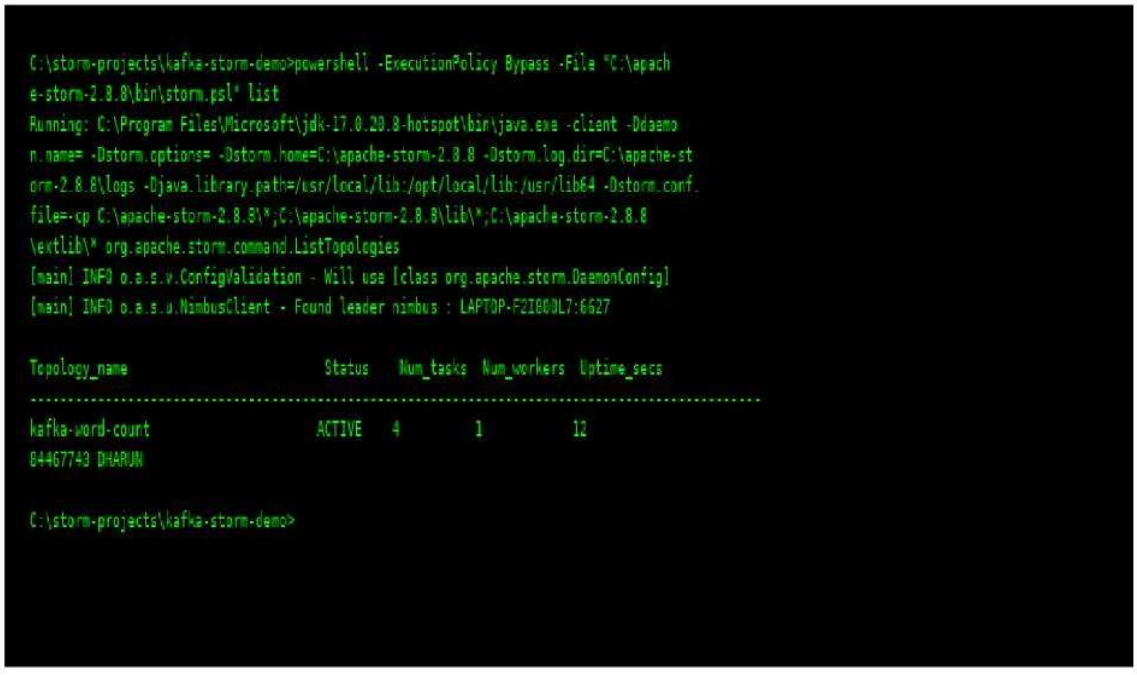


Figure 7.6: Topology Successfully Resubmitted and Running